THE UNIVERSAL PROPERTY OF THE MODULE OF DIFFERENTIALS: MAPS OUT ARE DERIVATIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The presentation of the module of differentials. Used in the counted proof at lines 34093-34096. Identifies the module as a quotient of the free module on the symbols, which is what the verification runs on.
- Derivations as rules with the Leibniz and linearity conditions. Used at the start of the counted proof (lines 34091-34092). Supplies the conditions that the induced map must satisfy.

Lemma 0.1. *The module of differentials of S over R has the following universal property. The map*

$$\mathrm{Hom}_S(\Omega_{S/R}, M) \longrightarrow Der_R(S, M), \quad \alpha \longmapsto \alpha \circ d$$

is an isomorphism of functors.

Proof. By definition an R-derivation is a rule which associates to each $a \in S$ an element $D(a) \in M$. Thus D gives rise to a map $[D] : \bigoplus S[a] \to M$. However, the conditions of being an R-derivation exactly mean that $[D]$ annihilates the image of the map in the displayed presentation of $\Omega_{S/R}$ above. □

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